Minglai Yang

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Working Experience

Scale AI — **San Francisco, CA**
Research Scientist, Agents — *Jul 2026 – Present*

Abaka AI — **Palo Alto, CA**
Senior Member of Technical Staff - Agents Lead — *Jan 2026 – Jun 2026*

- Led a 15-person cross-functional team on an MCP-based agent RL project, coordinating environment development, synthetic/real data generation, expert annotation, and agent evaluation workstreams.
- Designed environment simulators spanning B2B and B2C RL environments and deployed Harbor for systematic agent benchmarking and evaluation.
- Built a synthetic task and rubrics generation pipeline, producing over 3,000 high-quality, long-horizon tasks with automated evaluation criteria across 10 environments with ~800 tools in total.
- Led OmniDocBench-Pro [2], overseeing evaluation design, data selection, and human annotation pipeline (**NeurIPS In Submission**).
- Co-authored research on instruction-guided video editing evaluation [3] (**COLM In Submission**) and LLM error verifiability [4] (**COLM In Submission**).

Coretechs — **Kensington, MD**
Machine Learning Engineer Intern (Full Time) — *May 2024 – Aug 2024*

- Developed GPT-powered Slack bots with retrieval-augmented generation (RAG) and integrated real-time Slack chat into the official website using JavaScript, increasing active user engagement by 57%.
- Led end-to-end website development, from design to deployment, leveraging AWS (EC2, RDS, S3), Nginx, and Docker for scalable infrastructure.
- Conducted performance testing with Selenium, simulating 500+ users to ensure system reliability under high load.
- Received the *Best Intern Award* for outstanding contributions to the company's first GenAI project.

The University of Arizona, AI Core — **Tucson, AZ**
Lead AI Engineer (Part Time) — *Feb 2024 – Dec 2024*

- Built tool-calling AI agents with Retrieval-Augmented Generation (RAG) for internal knowledge; trained a custom long-context documentation QA model, improving top-1 accuracy by 48% on internal benchmarks.
- Designed and shipped an end-to-end campus chatbot website serving 50,000+ UA students with long-context retrieval; increased student web engagement by $5.39\times$ post-launch.

Research Experience

University of Arizona, CLU-LAB — **Tucson, AZ**
Undergraduate Research Assistant — *Oct 2024 – Dec 2025*

- Advisor: Prof. Mihai Surdeanu, Dr. Liangming Pan and Dr. Steven Bethard
- Led Research Project [1] on analyzing LLM reasoning, designed GSM-DC, a controlled benchmark for evaluating language models reasoning under irrelevant context, revealing **6 new empirical laws** on how irrelevant context disrupts reasoning, which training regimes improve robustness, and proposing test-time methods that achieve up to **6.3%** out-of-distribution performance gains (**EMNLP Oral – First Author**).
- Conducting research with Dr. Vikas Yadav on speculative decoding strategies for LLMs, introducing CopySpec [2] , a copy-and-verify approach that delivers up to **3.08× faster generation** and **2.66× higher efficiency** in LLM self-correction reasoning (**EMNLP Oral - Second Author**).
- Leading a project with Prof. Steven Bethard on long-context LLM reasoning, investigating whether model reasoning is trustworthy versus prone to hallucination and other failure modes (**SemEval In Submission - First Author**).
- Collaborated with Dr. Chicheng Zhang to develop LLM-guided warm starts for RL [3], achieving at least $20\times$ higher data efficiency (**EXAIT@ICML Workshop - Second Author**).
- Proposed using PPO to optimize the HIT@DICT reward for relation extraction; our RL optimization raised one-shot NYT29 F1 **from 24.65% to 48.11%**, with more stable convergence (**ICLR In Submission - Third Author**).

University of Arizona, ML4AI-LAB and IVI-LAB — **Tucson, AZ**
Undergraduate Research Assistant — *Oct 2023 – Dec 2025*

- Advisor: Prof. Kobus Barnard and Dr. Adarsh Pyarelal
- Developing dynamic Bayesian networks and Theory of Mind models within the ToMCAT project to improve AI modeling of human coordination.
- Leading a project on evaluating LLM planning capabilities under adversarial conditions, designing a controlled testbed to conduct stepwise analysis and benchmarking.
- Designing probabilistic models to disentangle causal interpersonal coordination from correlation.
- Co-authored a $544K, two-year research proposal, focusing on LLM planning, temporal-causal reasoning, and robustness evaluation against noisy commands.

Education

University of Arizona **Tucson**
B.S. in Computer Science *2023–2025*
- GPA: *4.0 / 4.0*
- Graduated December 2025 (Accelerated completion of 4-year curriculum in 2 years)
- All relevant coursework available here.

Publications

1. *How Is LLM Reasoning Distracted by Irrelevant Context? An Analysis Using a Controlled Benchmark*
   Minglai Yang, Ethan Huang, Liang Zhang, Mihai Surdeanu, William Wang, Liangming Pan
   EMNLP 2025 — Main Conference **[Oral]** (CORE A*), 2025 [preprint] [code] [slides] [poster] [project page]
2. *CopySpec: Accelerating LLMs with Speculative Copy-and-Paste Without Compromising Quality*
   Razvan Dumitru, **Minglai Yang**, Vikas Yadav, Mihai Surdeanu
   EMNLP 2025 — Main Conference **[Oral]** (CORE A*), 2025 [preprint] [code]
3. *Improving the Data-efficiency of Reinforcement Learning by Warm-starting with LLM*
   Thang Duong, **Minglai Yang**, Chicheng Zhang
   Exploration in AI Today (EXAIT) Workshop @ ICML, 2025 [preprint] [code]
4. *AlignSAE: Concept-Aligned Sparse Autoencoders*
   Minglai Yang, Xinyu Guo, Zhengliang Shi, Jinhe Bi, Steven Bethard, Mihai Surdeanu, Liangming Pan
   Transactions on Machine Learning Research (TMLR), 2026 [openreview] [preprint] [code] [project page]

Preprints

1. *Peeking inside the Black-Box: Reinforcement Learning for Explainable and Accurate Relation Extraction*
   Xinyu Guo, Zhengliang Shi, **Minglai Yang**, Mihai Surdeanu
   ICLR In Submission, 2026 [preprint][code]
2. *OmniDocBench-Pro*
   Minglai Yang, et al.
   NeurIPS In Submission, 2026
3. *VEFX-Bench: Video Editing and Effects Benchmark and Reward Model*
   Xiangbo Gao, Sicong Jiang, Bangya Liu, Xinghao Chen, **Minglai Yang**, Siyuan Yang, Mingyang Wu, Jiongze Yu, Qi Zheng, Haozhi Wang, Jiayi Zhang, Zihan Wang, Qing Yin, Zhengzhong Tu
   COLM In Submission, 2026
4. *Justified or Just Convincing? Error Verifiability as a Dimension of LLM Quality*
   Xiaoyuan Zhu, Kimberly Truong, Riccardo Fogliato, Gokul Swamy, Weijian Zhang, **Minglai Yang**, Longtian Ye, Bangya Liu, Minghao Liu, Andrew Ilyas, Steven Wu
   COLM In Submission, 2026

Honors and Awards

2025: **2nd Place — Reddit Wildcat Hackathon** *Reddit & University of Arizona*
2025: **Undergraduate Research Travel Grant (Top 10 awardees university-wide)**, *University of Arizona*
2025: **Galileo Circle Scholar (Top 0.8% Academic Award)**, College of Science, *University of Arizona*
2023–2025: **Academic Highest Distinction (Dean's List) (6 times)**, *University of Arizona*

2023–2025: **Global Wildcat Scholarship**, *University of Arizona*

2021: **Shanghai High School Science and Technology Competition**, Second Prize, *Shanghai High School*

2020: **Shanghai Youth Science and Technology Innovation Competition**, Third Prize, *Shanghai, China*

Leadership and Extracurricular Activities

AI Club, University of Arizona **Tucson, AZ**
President *Aug 2024 – Dec 2025*

- Advisor: Dr. Zhuolin Yang, Dr. Adarsh Pyarelal and Prof. Kobus Barnard
- Organized workshops on AI agents and LLM applications, including a Hack Arizona event with 100+ participants.
- Led weekly lectures and invited speaker sessions, engaging over 40 students across disciplines to foster AI learning.
- Organized a reading group that mentors students interested in AI research.
- Raised over $14,000 in sponsorships to support club activities and student research projects.

Mathematical Contest in Modeling (MCM), University of Arizona **Tucson, AZ**
Team Leader & First Author *Sep 2023 – Mar 2024*

- Led weekly tutoring sessions on mathematical modeling, convex optimization, Bayesian inference, and deep neural network methods.
- Developed mathematical models for the 2024 MCM problem on sea lamprey ecosystems, focusing on resource availability and sex ratio dynamics.

Teaching

Fall 2025: TA, CSC-244: Discrete Mathematics II, Computer Science, University of Arizona

Fall 2025: Instructor, Deep Learning and Natural Language Processing, AI Club, University of Arizona

Spring 2025: TA, CSC-144: Discrete Mathematics I, Computer Science, University of Arizona

Fall 2024 – Spring 2025: Instructor, Math for AI Workshop Series, AI Club, University of Arizona

Service

ACL Reviewer *2026*

ICML Reviewer *2026*

AI4MATH Workshop @ ICML Reviewer *2024*

Technical Skills

Languages & Systems: Python, Java, C/C++, JavaScript, MATLAB, Ruby, Swift; Linux, Git, Docker, AWS, Nginx

ML/DL & Finetuning: PyTorch, NumPy, pandas, scikit-learn, PyMC; PEFT (LoRA/QLoRA, Prefix/P-Tuning), SFT/DPO/PPO, RLHF/RLAIF; LLaMA-Factory, VERL, OpenRLHF, DeepSpeed-Chat; bitsandbytes

NLP & RAG/Serving: Hugging Face (Transformers/Datasets/Tokenizers), spaCy, NLTK, SentencePiece, Sentence-Transformers; LangChain, LlamaIndex, FAISS, Milvus, Qdrant, Weaviate, Chroma, pgvector; Elasticsearch/BM25, vLLM, FlashAttention, TransformerLens